22 $\left(a+\sqrt{8}\right)^2$ can be written in the form $c + d\sqrt{2}$, where a, c and d are integers.

Find, in terms of a, an expression for c **and** an expression for d.

$(a + \sqrt{8})^2 = a^2 + a\sqrt{8} + a\sqrt{8} + 8$

1 mark
All four products

$= a^2 + 8 + 2a\sqrt{8} = a^2 + 8 + 4a\sqrt{2}$

$c = a^2 + 8$

1 mark
$c=a^2+8$

$c =$ .....

$d = 4a$

1 mark
$d=4a$

$d =$ .....

(Total for Question 22 is 3 marks)